

Nolan Black, Ph.D. | Simulation Engineer

Redacted – U.S.A.

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A research engineer with a strong record in computational engineering, design optimization, and ML for physical simulation. Experienced in HPC programming for numerical analysis and python programming for ML and surrogate development.

Professional & Research Experience

Pasteur Labs | Scientific ML Startup (Remote) Philadelphia, PA

Simulation R&D Engineer January 2025 – Present

- Developed and executed CFD workflows (Ansys Fluent) to generate high-fidelity datasets supporting surrogate modeling for engineering design
- Designed differentiable software to perform structural optimization of pre-trained ML surrogates in aerospace applications (Python/PyTorch)
- Collaborated with cross-functional engineers to incorporate simulation results (from Ansys Fluent, OpenFOAM, Siemens StarCCM+) in iterative design workflows

Drexel University | GAANN Fellowship Philadelphia, PA

Graduate Research Assistant September 2020 – January 2025

- Developed and validated high-performance finite-element models for multiscale design optimization (C++/PETSc)
- Investigated novel methods of Machine Learning (ML) for physics-informed simulation (C++/Python/PyTorch)
- Integrated ML techniques to design nature-inspired metamaterials with tailorable physical properties

Lawrence-Livermore Nat. Lab. | Center for Design Optimization Livermore, CA

NSF PhD INTERN January 2024 – May 2024

- Executed high-performance computing workflows in a U.S. national laboratory environment
- Implemented parallel solvers for mixed finite element methods in a high-performance computing environment (C++)
- Developed nonlinear finite element programs for hyperelastic, hierarchical materials for applications in structural optimization

Zimmer Biomet | Zimmer Knee Creations R&D Exton, PA

Development Engineering Coop April 2019 – April 2020

- Developed functional medical devices using rapid prototyping to perform orthopedic procedures on the hand and wrist
- Coordinated with orthopedic surgeons to established design criteria and verify medical device designs in cadaveric lab testing
- Adapted, tested, and re-validated existing products to meet new standards of quality based on risk management

NAVSEA | Department of the Navy Philadelphia, PA

Entry-level Engineer April 2017 – September 2018

- Supported US Navy shipboard engineering systems through development and revision of procedures and system documentation
- Coordinated with shipboard engineering personnel to translate operational requirements into clear, auditable engineering documentation

Skills & Interests

Computational

- C++ (MPI, PETSc, mfem)
- Python (PyTorch, JAX, SciPy, NumPy)
- MATLAB

CAE Tools

- CFD: Ansys Fluent, OpenFOAM, StarCCM+
- Solid Mechanics: mfem, Ansys Mechanical
- CAD: Solidworks, Gmsh, CREO, Paraview

Select Publications

Black, Nolan and Najafi, Ahmad R. (2025), *Second-order homogenization in multiscale structural optimization.* Struct. Multidiscip. Optim., Manuscript accepted for publication.

Black, Nolan and Najafi, Ahmad R. (2024), *Stress-constrained optimization of multiscale structures with parameterized microarchitectures using machine learning.* Struct. Multidiscip. Optim., 10.1007/s00158-024-03821-y.

Black, Nolan and Najafi, Ahmad R. (2023), *Deep neural networks for parameterized homogenization in concurrent multiscale structural optimization*. Struct. Multidiscip. Optim., 10.1007/s00158-022-03471-y.

Black, Nolan and Najafi, Ahmad R. (2022), *Learning finite element convergence with the Multi-fidelity Graph Neural Network*. Comput. Methods Appl. Mech. Eng., 10.1016/j.cma.2022.115120.

Black, Nolan and Najafi, Ahmad R., *Neural networks for nonlinear homogenization in multiscale structural optimization*. Manuscript under peer review.

Select Conference Presentations

Black, Nolan and Najafi, Ahmad R., "Multiscale structural optimization with strain gradient effects using second order homogenization". World Congress on Computational Mechanics (WCCM2024), Vancouver, BC, July 21-26, 2024.

Black, Nolan and Najafi, Ahmad R., "Second-order Homogenization for Structural Optimization Applications". Engineering Mechanics Institute Conference 2024 (EMI2024), Chicago, IL, May 28-31, 2024.

Black, Nolan and Najafi, Ahmad R., "Multiscale Design Optimization with Local Failure Criteria through Machine Learning". 17th U.S. National Congress on Computational Mechanics (USNCCM17), Albuquerque, NM, July 23-27, 2023.

Black, Nolan and Najafi, Ahmad R., "Neural Network Surrogates for Multiscale Structural Optimization". 2022 International Mechanical Engineering Congress and Exposition (IMECE 2022), Columbus, OH, Oct. 30 – Nov. 02, 2022.

Black, Nolan and Najafi, Ahmad R., "Multi-Fidelity Graph Neural Networks as Surrogate Models for Finite Element Analysis". 19th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM19), Austin, TX, June 19-24, 2022.

Black, Nolan and Najafi, Ahmad R., "Deep Learning in Concurrent Multiscale Structural Optimization". 19th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM19), Austin, TX, June 19-24, 2022.

Black, Nolan and Najafi, Ahmad R., "Incorporating Architected Materials in Multiscale Structural Optimization with Deep Learning". Engineering Mechanics Institute Conference 2022 (EMI2022), Baltimore, MD, May 31-June 3, 2022.

Scholarships, Fellowships, and Honors

- NSF Non-academic Research Internship (INTERN) Award (2024)
- Koerner Family Foundation Fellowship (2022)
- Harry L. Brown, Jr. Memorial Fellowship (2021)
- Fellow, U.S. E.D. Graduate Assistance in Areas of National Need (2020-Present)
- A.J. Drexel Scholarship, Drexel University (2016–2020)

Professional Affiliations and Service

- Member, The American Society of Mechanical Engineers (2016–Present)
- Student Representative, Graduate Studies Committee, Drexel University (2022–2023)
- Entrepreneurial Lead, NSF I-Corp (2022–2023)

Teaching and Mentorship

MEM592

Applied Engineering Analytical & Numerical Methods II, Drexel University

Co-teacher and Teaching Assistant

Graduate Research Mentor

Drexel University

Served as primary mentor for one M.S. thesis.

Undergraduate Research Mentor

Drexel University

Served as primary mentor for three undergraduate researchers.

Education

Ph.D., Mechanical Engineering / Computer Science

Drexel University, Philadelphia, PA

September 2020 – December 2024

Thesis: Multiscale Structural Optimization through Second-order Homogenization and Machine Learning

B.S./M.S., Mechanical Engineering

Drexel University, Philadelphia, PA

September 2016 – June 2020